

NITYA MITTAL

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WORK EXPERIENCE

American Express | *Software Engineer - Associate DPM*

July 2023 - July 2024

- **Saved ~\$70K** in Q1-Q2'24 through **33% data rationalization**, by designing & developing **Python** and **SQL** pipelines to resolve data quality issues and implementing optimization strategies for the **Hive** data layers used by the Anti-Money Laundering Transaction Monitoring team, Global Risk & Compliance Dept.
- Led the implementation of onboarding pipelines for Kabbage acquisition, **migrating ~20K accounts** into Amex's banking ecosystem through scalable ingestion, validation, and reconciliation frameworks that ensured AML compliance.
- Maintained and debugged production pipelines ingesting **millions of transactions** per 5-minute Amex feed, detecting file loss and duplication through transaction-volume anomaly monitoring in **Yellowbrick**; partnered with the AML Transaction Monitoring team via **JIRA** in an Agile environment to triage and resolve data-integrity issues.

American Express | *Data Engineering - Analyst Intern*

May 2022 - July 2022

- Reduced development time by 80% for Model Performance Tracking of the Expected Credit Loss Forecasting Tool, by building **Python** scripts in **MLStudio** to consolidate all US and global portfolios at the Credit & Fraud Risk Dept.
- Automated data sourcing across multiple formats (SQL, JSON) by designing object-oriented **Java** services with secure coding principles to pull and load data into **Hive**, following SDLC.
- Streamlined onboarding for new data engineers by authoring technical design documentation in **Confluence** detailing BAU and ad-hoc workflows, sharing knowledge to support the team.

TECHNICAL SKILLS

Language: Python, C, C++, R, SQL, MATLAB, TypeScript, JavaScript

Cloud Tech: Amazon Web Services; Google Cloud : GCP, Vertex AI, Firestore.

DevOps, Infra & Delivery: Git, Docker, Kubernetes, Linux OS, SDLC, CI/CD.

App Dev & Analytics: HTML, CSS, REST APIs, React, Node.js, FastAPI, Uvicorn, Streamlit, n8n, PHPUnit.

ML & AI: scikit-learn, numpy, scipy, pandas, LangChain, keras, tensorflow, LangGraph, transformers; BERT, Gemini 2, RoBERTa, all-MiniLM-L6-v2, all-mpnet-base-v2, Opacus, Presidio, NetworkX.

AI Coding Tools: Claude Code, Codex, Cursor, Github Copilot

EDUCATION

University of Michigan, Ann Arbor

Aug. 2024 – May 2026

Master of Science in Engineering in Computer Science

GPA: 4.0/4.0

Delhi Technological University, Delhi

July 2019 – May 2023

Bachelor of Technology in Mathematics & Computer Science

GPA: 3.81/4.0

Coursework: Machine Learning, Natural Language Processing, Data Science, Statistics, Linear Algebra, DSA, DBMS, System Design.

PUBLICATION

“**University Recommendation System**” presented at **International Research Conference on Science Technology, Engineering and Management (IRCSTEM)** on 6–7th May 2023 at Gurgaon, India.

PROJECT EXPERIENCE

Needle: RAG-based Research Assistant

Aug 2025 - Nov 2025

- Built an RA for literature discovery & source-grounded Q&A over **1.7M+ arXiv papers** & uploaded PDFs, removing hallucinated citations through retrieval-backed generation. Developed a RAG pipeline using **Streamlit**, **Vertex AI**, **Gemini 2**, & **LangChain** with semantic search, citation verification, & conversational document querying.

CineRank: Real-Time Movie Recommendation System

Aug 2025 - Oct 2025

- Built a collaborative-filtering recommender with **TruncatedSVD** (50 latent factors) over a **200K-user × 87K-movie** matrix from **MovieLens 32M**, using cold-start fallback to popularity-ranked titles so the system never returns empty results. Served recommendations at **<100ms latency** through a streaming pipeline of **Kafka**, **Redis**, **PostgreSQL**, & **FastAPI** handling **1000+ events/sec**.

PrivNLGBench: Differential Privacy Benchmark

Jan 2025 - Jun 2025

- Built a benchmark for evaluating privacy-preserving NLP systems by transforming real-world datasets (Yelp, Enron, Clinton FOIA) into structured test sets with isolated PII and PII-free summaries. Developed the evaluation pipeline using **PyTorch**, **Opacus**, **DistilRoBERTa**, & **NetworkX**, achieving **0.85 accuracy** / **0.71 macro-F1** for PII detection and improving calibration from **0.272 to 0.083 ECE**.

MiniGPT: From-Scratch GPT-Style Language Model

Sep 2025 - Oct 2025

- Implemented a decoder-only GPT-style **transformer from scratch** in **PyTorch** (multi-head causal self-attention, positional embeddings, & residual blocks), pretrained on **~1B tokens** of **FineWeb-Edu** via next-token prediction. Applied supervised fine-tuning on **SmolTalk** multi-turn dialogues with assistant-only loss masking, then evaluated on a multiple-choice QA benchmark.

ACHIEVEMENTS, EXTRACURRICULARS & LEADERSHIP

- **Awards:** McKinsey Next Gen Women Leaders 2023 & 2025 | **URAM (Micron) Scholarship** 2022 | **AWS ML Scholarship** 2021
- **Leadership:** Judge, **MHacks 17 (MLH Official)** | **Vice Chairperson**, IEEE DTU CS | Networking Officer, GradSWE
- **Mentorship:** Mentored 300+ students at IEEE DTU, JA India & AmEx, Joy of Coding (UMich)